

# Solution Architecture

Project: Rising Waters – Flood Prediction using Machine Learning

Team Lead: Atluri Jyothendra Sai Ram

Members:

- Battu Rama Rao
- Hima Varshini Bodapati
- Siddhardha Jilagam
- Udaya Teja

## Purpose

This document explains the 'Solution Architecture' stage of the Rising Waters project. The project predicts flood occurrence using historical weather data and machine learning. The documentation follows the SDLC flow from ideation to deployment.

## Project Overview

Project Overview for this phase focuses on developing a reliable flood prediction solution. Historical rainfall, seasonal rainfall, and cloud visibility are analyzed to train classification models. This phase contributes to improving early warning, evacuation planning, disaster management, and decision making.

## Objectives

Objectives for this phase focuses on developing a reliable flood prediction solution. Historical rainfall, seasonal rainfall, and cloud visibility are analyzed to train classification models. This phase contributes to improving early warning, evacuation planning, disaster management, and decision making.

## Dataset Description

Dataset Description for this phase focuses on developing a reliable flood prediction solution. Historical rainfall, seasonal rainfall, and cloud visibility are analyzed to train classification models. This phase contributes to improving early warning, evacuation planning, disaster management, and decision making.

## Machine Learning Models Used (Decision Tree, Random Forest, KNN, XGBoost)

Machine Learning Models Used (Decision Tree, Random Forest, KNN, XGBoost) for this phase focuses on developing a reliable flood prediction solution. Historical rainfall, seasonal rainfall, and cloud visibility are analyzed to train classification models. This phase contributes to improving early warning, evacuation planning, disaster management, and decision making.

## **Detailed Discussion**

### **Flask Web Application**

The Flask application integrates the best-performing model (XGBoost achieving about 96.55% accuracy). The application provides an easy interface for authorities to enter weather parameters and instantly receive flood risk predictions. Deployment on IBM Cloud improves scalability, accessibility, and future maintenance.

### **IBM Cloud Deployment Plan**

The Flask application integrates the best-performing model (XGBoost achieving about 96.55% accuracy). The application provides an easy interface for authorities to enter weather parameters and instantly receive flood risk predictions. Deployment on IBM Cloud improves scalability, accessibility, and future maintenance.

### **Expected Outcomes**

The Flask application integrates the best-performing model (XGBoost achieving about 96.55% accuracy). The application provides an easy interface for authorities to enter weather parameters and instantly receive flood risk predictions. Deployment on IBM Cloud improves scalability, accessibility, and future maintenance.

# Team Contributions & Summary

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- Siddhardha Jilagam
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## Responsibilities

Team Lead coordinated planning, integration, and reviews. Battu Rama Rao contributed to documentation, backend integration, and testing. Hima Varshini Bodapati assisted with requirement analysis and UI documentation. Siddhardha Jilagam supported model evaluation and testing. Udaya Teja worked on dataset preparation and deployment support.

## Conclusion

The 'Solution Architecture' document forms an important part of the complete project lifecycle. Together with the remaining documents it provides end-to-end evidence of planning, implementation, testing, documentation, and demonstration.